

NSF PESOSE Info Session

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OSPO@GT Core Objectives – 2025-2027

OSPO Goal 1: Support GT OSPO Sustainability

Improve OSS Developer Engagement

Create a welcoming community for all developers and engage in robust community outreach

Develop added metrics to track OSPO progress and criteria for success

Support Developers at GT and other Atlanta universities through workshops, training sessions, and meet-ups

Institutionalize Student Programs

Evolve the virtual summer internship program to be more sustainable and allow for more opportunity for all interested students

Enhance and Improve training materials to allow for more large-scale, asynchronous OSS training programs

Address the AI Revolution

Educate students on the benefits and pitfalls of open-source AI and current viewpoints, including the ethical use of AI

Host Open Data Sets and exploration tools for AI data agents

Provide training for AI hardware and software tools on local HPC resources

Design joint software engineering training with other GT groups

Explore GT Strengths in OSS

Collaborate with Georgia Tech's School of Cybersecurity and GTRI to develop and share secure open-source best practices.

Support institute efforts in maintaining research integrity using open-source techniques

OSPO Goal 2: Support a Sustainable OSS Community for GT and Atlanta

About Session

- NSF PESOSE – What has changed?
- Target Applicants
- Submission Requirements
 - Track I versus Track II submissions
- OSPO Resources for PESOSE applicants
- Other possible sources for funding

Note that this presentation is meant to help supplement existing NSF resources. We strongly encourage you to review all NSF resources and solicitations as the definitive guide for proposals.

<https://www.nsf.gov/funding/opportunities/pesose-pathways-enable-secure-open-source-ecosystems>

About NSF PESOSE (26-506)

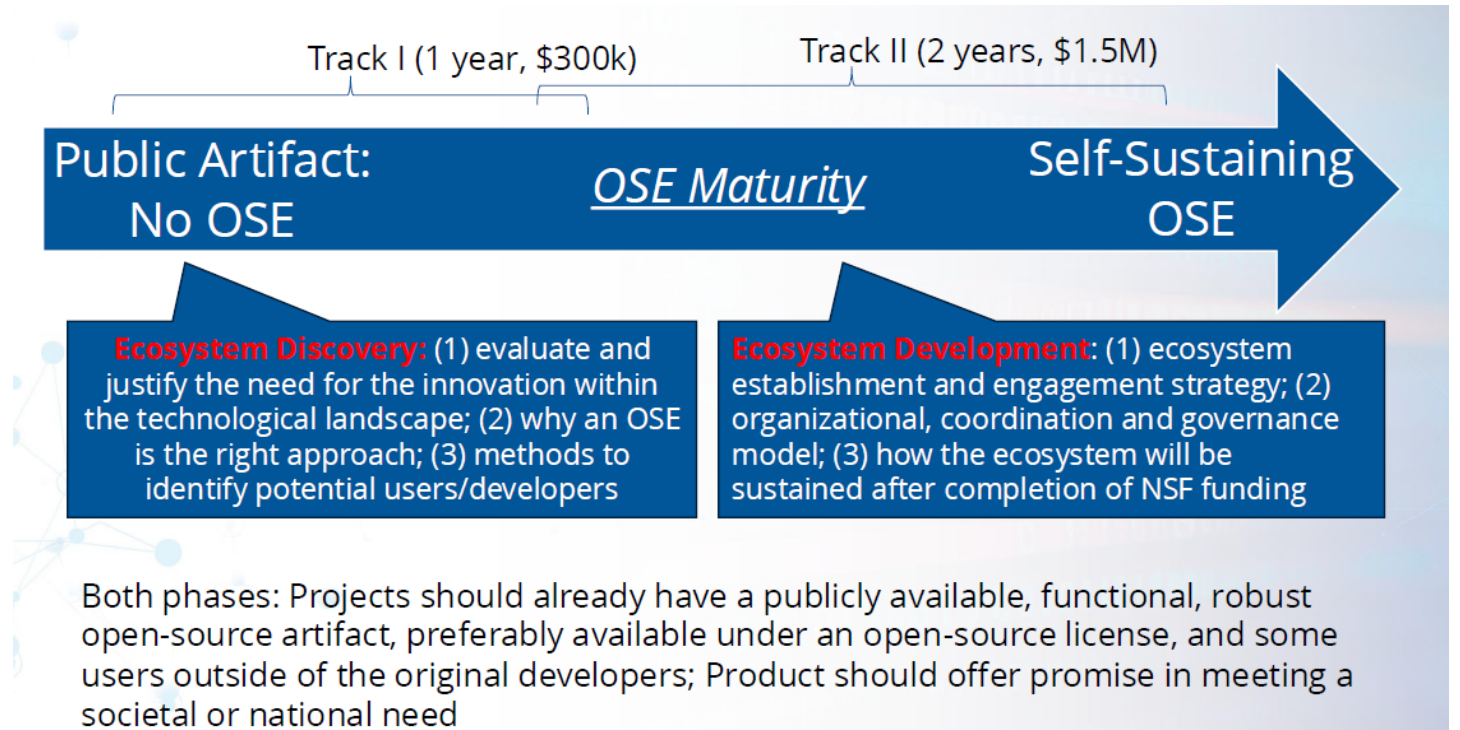
- Call for Pathway to Enable Secure Open-Source Ecosystems (PESOSE) proposal solicitation [[link](#)]
(all pictures in this slides deck are from [this link](#))

Due Dates: 9/1/2026 and 3/2/2027

- Vision
 - Forming managing organizations
 - Ensuring a broader and diverse adoption of open-source products
 - Enabling collaboration and catalyzing further innovation
 - Focusing on technologies with broad societal impact
- OSEs must use a distributed development model (e.g., GitHub, GitLab)
- PESOSE program **does not**
 - Support **development of** open-source research products
 - Including new core features
 - Support existing **well-resourced** open-source communities and ecosystems
 - Support **open-source ventures** for which **profit is the primary motive**

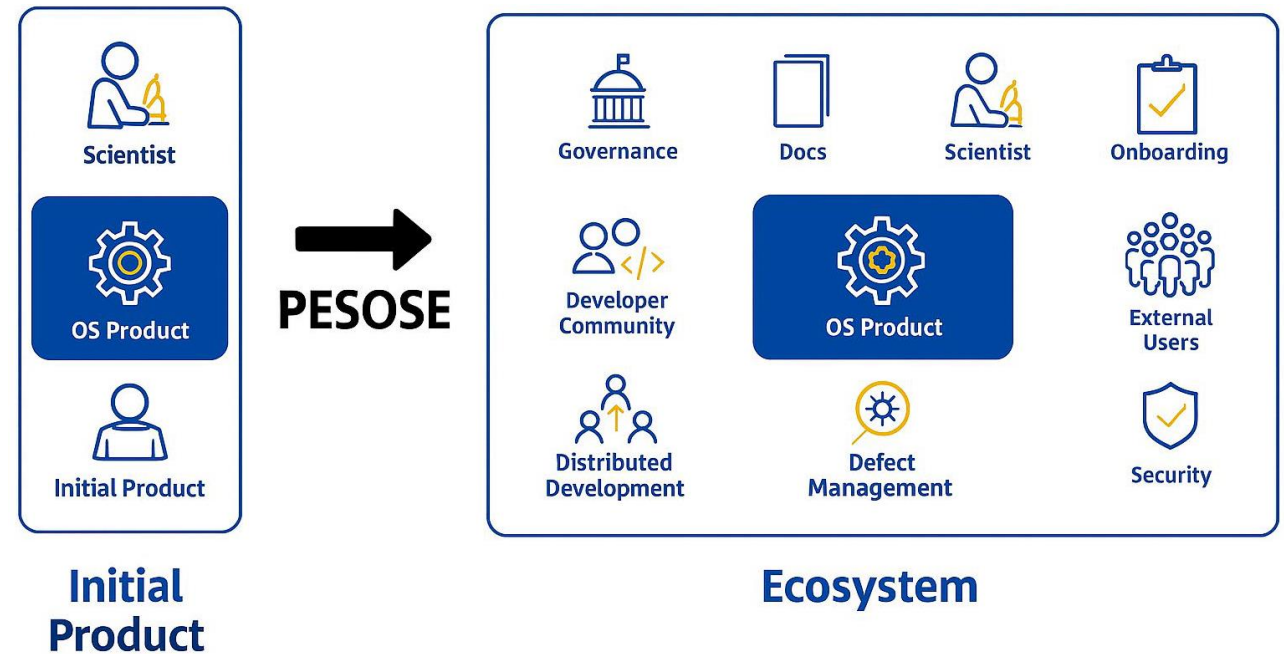
Target Applicants

- Have an existing open-source product/class of products
- Have active users outside of the founding team
- Need to build open-source ecosystems
- Willing to take responsibility for creating and managing the processes and infrastructure

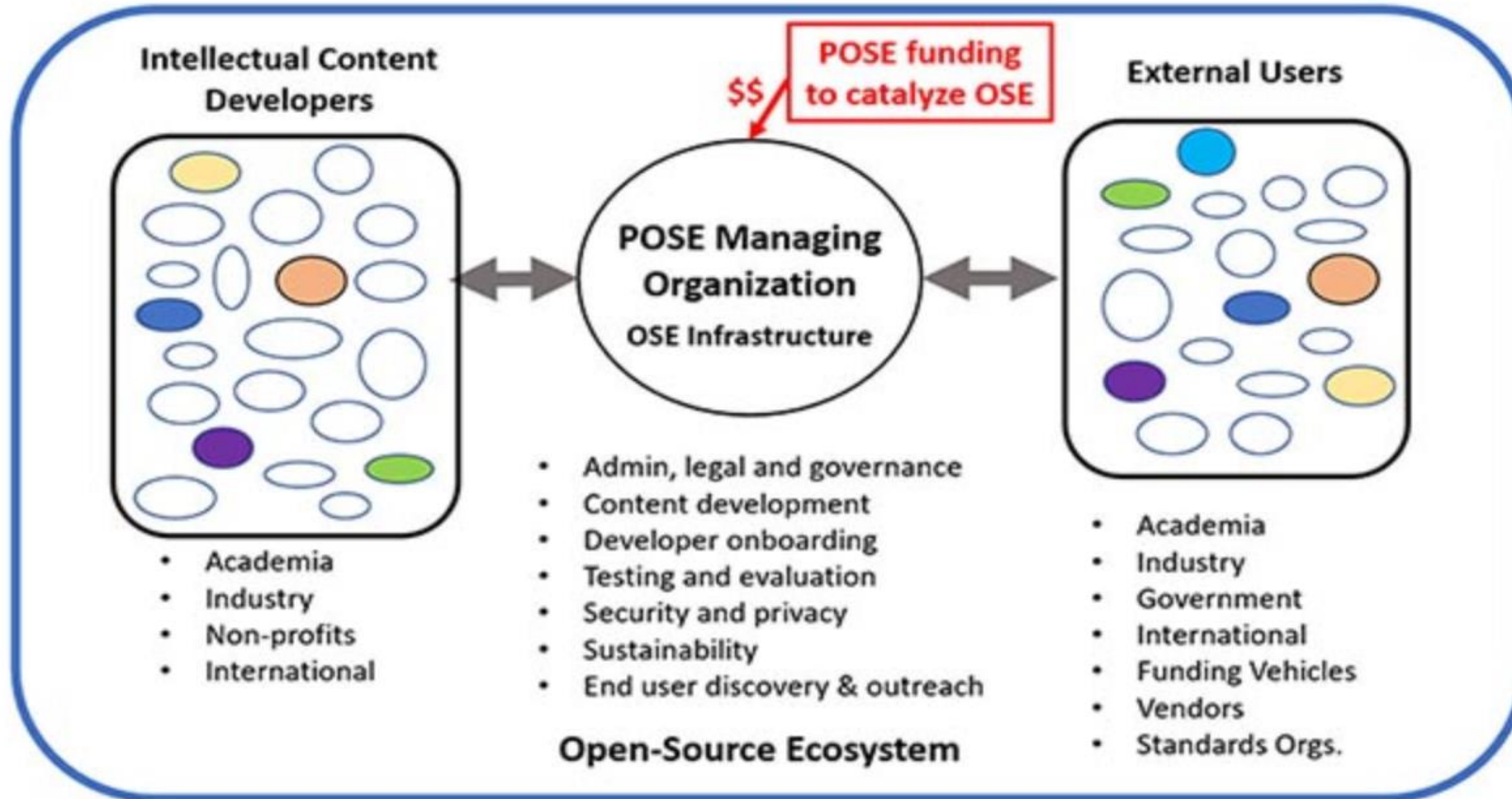


What is an Open-Source Ecosystem (OSE)?

- A self-sustaining **community** organization
- Enables ongoing, distributed, asynchronous development of an open-source product
- Uses an open-source licensing model
- Key components:
 - A leadership team for governance and management
 - A decentralized and open network of intellectual content developers
 - Users whose needs serve to guide the evolution of the product
- A key aspect of successful OSEs is their value in demonstrating the potential of a technology at scale in diverse application scenarios.



Role of the P(ES)OSE Managing Organization



PESOSE Guidelines

- Track I (\$300,000 up to one year):
 - **Scope and plan for an OSE** (ecosystem discovery, organization and governance, community building)
 - Up to 7 pages project description
- Track II (\$1,500,000 up to two years):
 - **Establish and expand a sustainable OSE** (Ecosystem Establishment/Growth, Organization and Governance, Continuous Development Model, Community Building, Sustainability)
 - Up to 15 pages project description
- Track III (\$1,500,000 up to two years):
 - **Improving safety, security, and privacy** (Addressing the key vulnerabilities of an OSE)
 - Up to 15 pages project description
- The primary distinction between Track I and Track II:
 - Track I's scoping and planning projects are intended for organizations that are less experienced or knowledgeable about their user base
 - Track II focuses on the development of non-technical roles within an open source project; e.g. corporate governance, legal and administrative functions, licensing, etc
 - Track II must have a **plan for sustainability** of the project after two years of PESOSE funding.

Estimated Number of Awards: 40-60 (30 Track I, 10 Track II, 10 Track III)

What Changed since 2025?

- PESOSE combines the old POSE [[link](#)] (Track I/II projects) with Safe-OSE [[link](#)]
 - Safe-OSE is still considered under Track III
 - Numbers of awards and funding levels change slightly
 - No limits on Track III awards – Safe-OSE had strict institutional limits on proposals
- New Deadlines
 - March 2027 deadline includes all tracks instead of just Track I
- Dear Colleague Letter (NSF 26-015) notes that NSF is interested in projects that support **open, AI agent ecosystems**

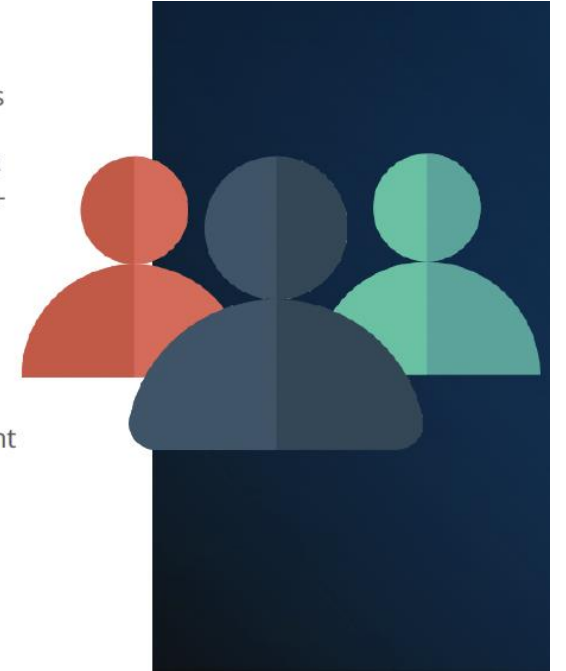
- DCL 26-015: <https://www.nsf.gov/funding/opportunities/dcl-advancing-artificial-intelligence-ai-agent-ecosystems>

PESOSE Community Discovery Component

PESOSE now requires a “I-Corps for PESOSE” style interview process to identify your OSE’s userbase

- 6 week required program in Fall/Winter
- 100 interviews required over the 4 weeks with a final report out
- Normally required for both Track I and Track II
- ***Consider who will be your community manager and industry mentor for any proposal!***

- 3-person team
 - **Team Lead** – Conducts most customer discovery interviews and gives progress reports at each meeting. We suggest that this might be a person who is regularly involved in communications with your users or stakeholders. Consider a graduate student/post-doc for this leadership role.
 - **Community Manager** – Jointly responsible for interviews and data collection/submission and should have an extensive understanding of the solution's potential user base. Should be the PI or co-PI.
 - **Industry Mentor** – Firsthand OSE experience to help guide efforts; should be independent of the technical development
- All team members must attend all sessions, engage in interviews
- Cameras on, prepared to fully engage
- 15+ interviews per week – 60+ total
- Weekly report-outs to present key learnings



Some of the Instructions from NSF POSE July 2024 Training Session

Note that NSF has also shared a DCL with more information for this effort as of March 2026.

<https://www.nsf.gov/funding/opportunities/pesose-pathways-enable-secure-open-source-ecosystems/updates/120507>

Recent GT PESOSE Projects – Track 1

Recent Track 1 awardees at GT include:

[EarSketch](#) – 2024

- PIs: Stephen Garrett, Brian Magerko, Jason A Freeman

[QCSchema/cclib](#) - 2025

- PIs/: David Sherrill, Lori Burns

[ML-Driven High-Level Synthesis](#) – 2026

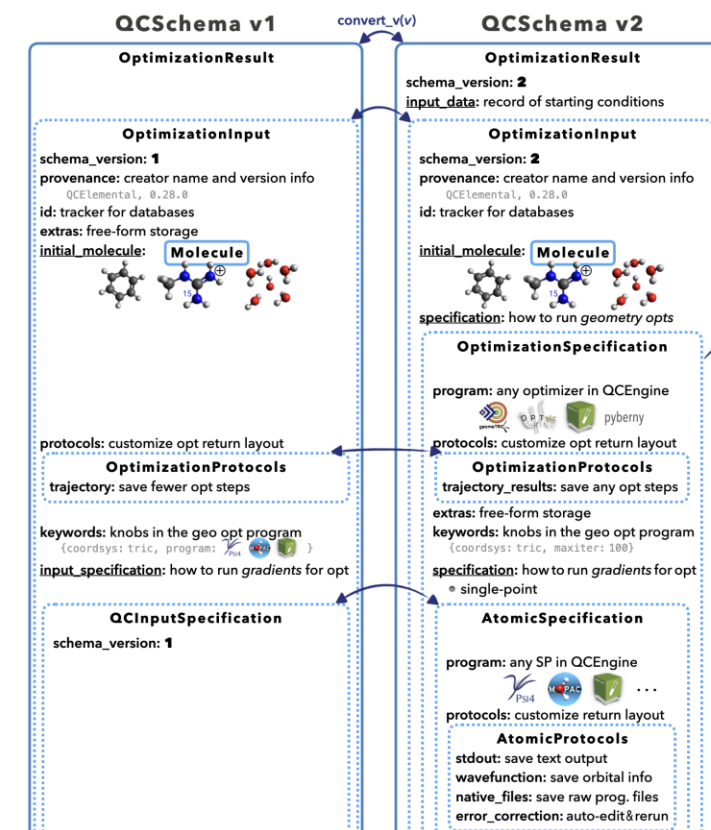
- PI: Callie Hao

[Accelerating the Sparse Array Programming Ecosystem](#) – 2026

- PIs: Willow Ahrens

[Vortex](#) (Open Source GPU) – 2026

- PIs: Hyesoon Kim, Jeff Young



Recent Updates to QCSchema

Recent GT PESOSE Projects – Track 2

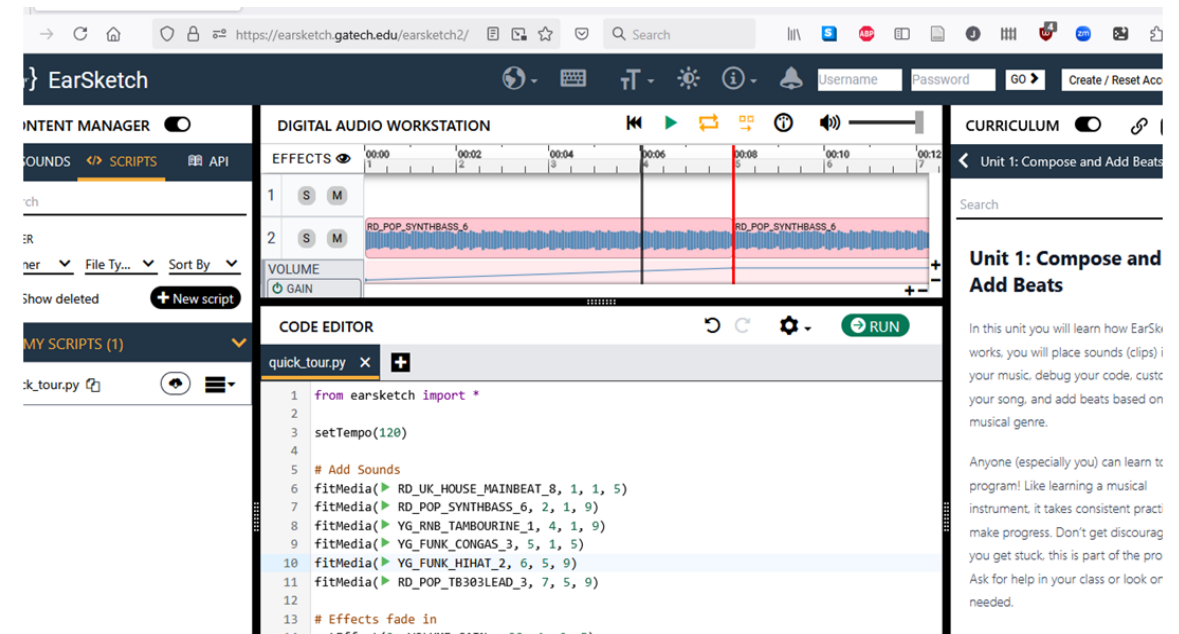
Recent Track 2 awardees at GT include:

[EarSketch](#) – 2026

- Pls: Stephen Garrett, Brian Magerko, Jason A Freeman

[Airavata Custos](#) – 2025

- Pls: Suresh Marru, Eroma Abeysinghe, Lahiru Jayathilake, Dimuthu Wannipurage



Earsketch Interface example

Stephen Garrett of Earsketch presented at the 2025 OSPO workshop and shared insights from their Track 1 experience

[\[Figshare Link to Slides\]](#)

PESOSE Track III

Track III is focused on supporting **mature** open source ecosystems, with a focus on safety, security, and privacy of these OSE organizations

- *Projects that are not yet at the OSE stage (target of the POSE Track 2) are likely not eligible*

Differences with Track I/II

- Focuses mostly on improving the security of an existing open source ecosystem rather than building an OSE
 - Proposers must clearly define the **vulnerability landscape** and how the PESOSE effort addresses **critical vulnerabilities** for an OSE
- Does not require I-Corps program

Similarities with Track I/II

- Does not support research or feature development for OSS
 - Focus is on improving an OSS organization and codebase
- Similar proposal requirements as Track II
 - E.g., 5 letters are highly suggested for both

Common FAQs

- Can I do any development that supports the OSE (ie, a new “feature”)?
 - The NSF webinar indicated that reasonable justification includes fixing security vulnerabilities in the OSE or features that are specifically needed to grow the community (e.g. a new tutorial or getting started framework)
 - Allocating more than 15-20% time for these efforts is likely to draw scrutiny from reviewers
- What does PESOSE mean by sustainability?
 - Not necessarily financial – that a project is drawing new contributors, users, and resources

OSPPO And GT Resources

GT's OSPPO can support your proposal in the following ways:

- Through programs like the OSPPO's [Virtual Summer Internship Program](#)
 - Student engagement can be budgeted to support developing OSE features and community engagement
- Collaboration letters
 - This letter should supplement other users of the OSE – the OSPPO is likely not a core contributor to your codebase!
 - Letters for this program do not have to follow the “generic” NSF format but should indicate the role of the collaborator as **third-party users or contributors to the open-source product**
 - It's highly suggested to have **5 letters**, especially for Track II proposals
- Proposal description paragraph defining the OSPPO's role and support of your project
- Pointers to useful tools and resources to improve your open source project in terms of licensing and software best practices
- OSPPO@GT can also help with identifying industry mentors for community engagement.
 - ***In some cases, OSPPO personnel can serve as the industry mentor for the I-Corps training program.***

Sample Text for NSF Proposal (can be tweaked)

The Georgia Tech Open Source Program Office (OSPPO) provides a central resource for training, best practices, and advice on how to grow and support open source projects and ecosystems. In the context of this proposal, the GT OSPPO will work with the PI to provide advice on how to best evolve their project towards a sustainable open source ecosystem through training seminars, advice on licensing and open source best practices, and tools and templates meant to help improve the quality of new and existing open source repositories.

Useful Software Resources You May Consider

RepoAuditor tool from Scientific Software Engineering Center

<https://github.com/gt-sse-center/repoAuditor>

RepoLinter from TodoGroup

<https://github.com/todogroup/repolinter>

Repo Security Audit Scorecards

<https://securityscorecards.dev/>

Templates for creating Contributing Guide

<https://github.com/nayafia/contributing-template>

Governance model explainers

<https://github.com/gt-ospo/oss-training/blob/main/notebook-lessons/oss-module-06-governance-foundations.md>

- Points to additional resources for selecting a governance model

Repo Scaffolding Resource

<https://dsacms.github.io/repo-scaffolder/>

Introduction

Write something nice here!

First off, thank you for considering contributing to Active Admin. It's people like you that make Active Admin such a great tool.

[source: [Active Admin](#)] Need more inspiration? [1] [Read The Docs](#) [2] [Mustache.js](#)

Tell them why they should read your guidelines.

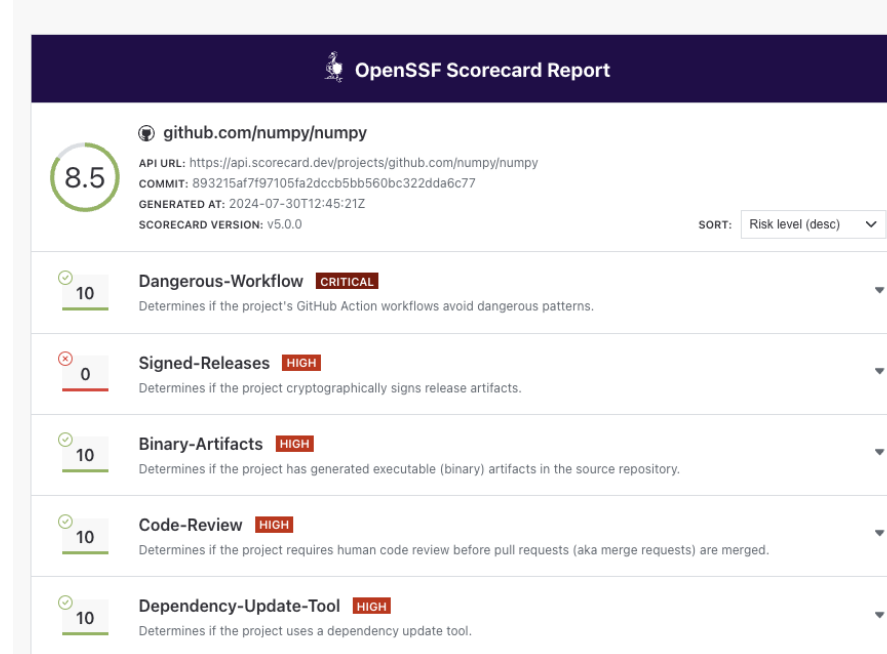
Following these guidelines helps to communicate that you respect the time of the developers managing and developing this open source project. In return, they should reciprocate that respect in addressing your issue, assessing changes, and helping you finalize your pull requests.

[source: [Hoodie](#)]

Explain what kinds of contributions you are looking for.

Keep an open mind! Improving documentation, bug triaging, or writing tutorials are all examples of helpful contributions that mean less work for you.

Elasticsearch is an open source project and we love to receive contributions from our community — you! There are many ways to contribute, from writing tutorials or blog posts, improving the documentation, submitting bug reports and feature requests or writing code which can be incorporated into Elasticsearch itself.



Useful Training and OSE Resource

Books

[Open Source Projects – Beyond Code](#)

- Covers governance models and the full open source lifecycle
- Available for free from GT Library via O'Reilley

[Working in Public](#)

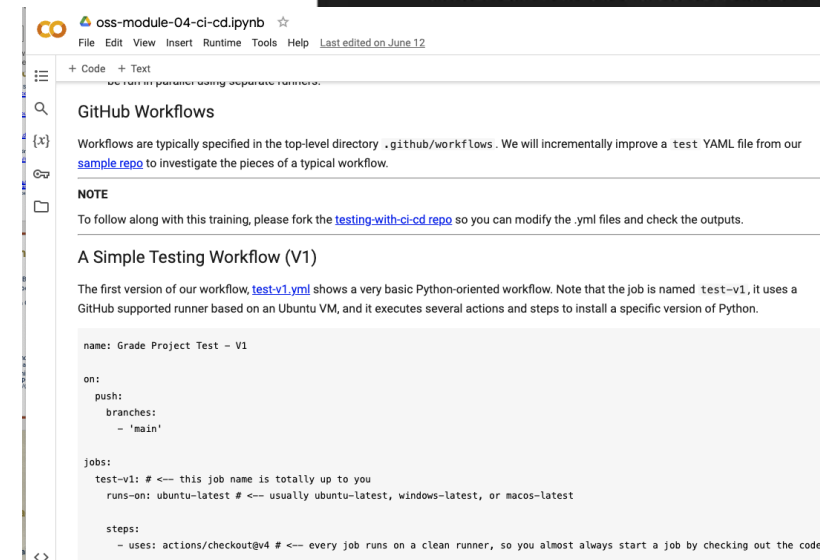
- Recent book focusing on GitHub and open source developer interviews

[It Takes a Village](#)

- A free guidebook focused on developing sustainability for cultural and scientific heritage organizations

OSS Training

- GT OSPO training is under development
 - Currently covers git, unit testing, CI/CD, code review, governance models, open source project best practices
 - <https://github.com/gt-ospo/oss-training>



Other Sources of Funding for OSS

Open Source for Science Fund [\[Link\]](#)

- Supports scientific projects with support for bioscience related efforts
- Seems to support similar projects as the former Chan Zuckerberg Initiative, “Essential Open Source Software for Science”

Apereo Foundation [\[Link\]](#)

- Supports a grant prioritization framework and an incubation program for scaffolding of new projects

Better Scientific Software Fellowships [\[Link\]](#)

- US based call for scientific software funding for workshop, tutorials, and tools that engage the scientific computing community

Research Software Alliance [\[Link\]](#)

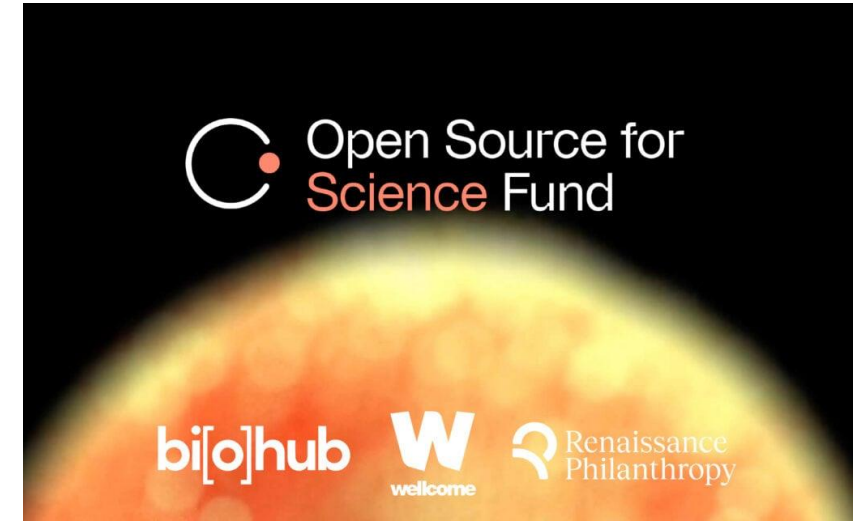
- Has an updated list of related calls

GitHub Secure Open Source Fund [\[Link\]](#)

- Supports critical infrastructure and keystone OSS projects

GT OSPO grants and Virtual Summer Internship Program [\[Link\]](#)

CURIOSS Funding Opportunities Dashboard [\[Link\]](#)





Reach out to us with questions!

Email: ospo-directors@groups.gatech.edu